



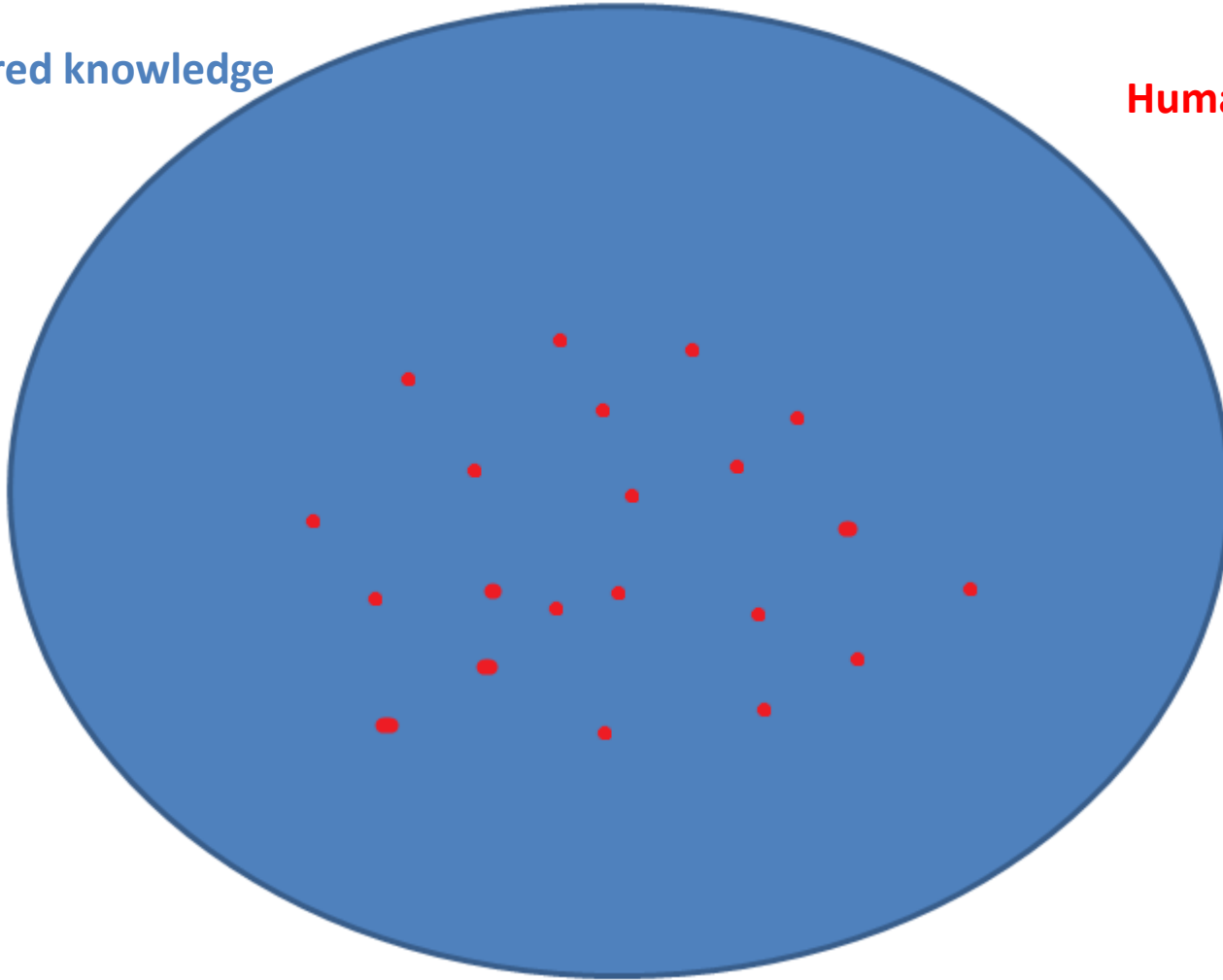
How to write a scientific Paper

Prof. Dr. Ahmad Hassanat
Faculty of IT, Mutah University

Filling the gap

Undiscovered knowledge

Human knowledge



**Read, Read, Read then red so you can change the blue into red,
then you can write , And tell the world to update Science**

Conveying the idea

- Here is a problem
- It's an interesting problem
- It's an unsolved problem
- **Here is my idea**
- My idea works (details, data)
- Here's how my idea compares to other people's approaches

Paper Structure

- Title, Name and affiliation
- Abstract (4 sentences)
- Introduction (1 page)
- Related work (1-2 pages)
- The problem (1 page), My idea (2 pages), The details (5 pages)
- Results and discussion (evidence to support claims)
- Conclusions and future work (0.5 pages)
- Acknowledgement
- References

Title

- a good title does several things:
 - First, it predicts content.
 - Second, it catches the reader's interest.
 - Third, it reflects the tone of the piece of writing.
 - Fourth, it contains keywords that will make it easy to access by a computer search

Title example

Solving the Problem of the K Parameter in KNN Classifier Using Ensemble Learning Approach

Ahmad B. A. Hassanat¹, Mohammad Ali Abbadi², Ahmad Ali Alhasanat³

^{1 2}IT Department, ^{1 2} Mu'tah University, Mu'tah – Karak, Jordan, 61710,

³ Al-Hussein Bin Talal University, Maan, Jordan

Email: ¹ahmad.hassanat@gmail.com, ² abbadi@mutah.edu.jo,

The abstract

- We usually write the abstract last
- Used by program committee members to decide which papers to read
- Four sentences:
 - 1. State the problem (background)
 - 2. what is your solution (methods)
 - 3. Say what your solution achieves (results)
 - 4. Say what follows from your solution (conclusions)
- Most journals limit it by 150-400 words maximum

Abstract example

This paper presents a new solution for choosing the K parameter in the k -nearest neighbor (KNN) algorithm, the solution depends on the idea of ensemble learning, in which weak a KNN classifier is used each time with a different K , starting from one to the square root of the size of the training set. The results of the weak classifiers are combined using weighted sum rule.

The proposed solution was tested and compared to other solutions using a group of experiments on real world problems. The experimental results show that the proposed classifier outperforms the traditional KNN classifier that uses different number of neighbors, is competitive to other classifiers, and is a promising classifier with strong potential for a wide range of applications.

Structured Abstract

Decision tree-based learning and laboratory data mining: an efficient approach to amebiasis testing

Abstract

Background Amebiasis represents a significant global health concern. This is especially evident in developing countries, where infections are more common. The primary diagnostic method in laboratories involves the microscopy of stool samples. However, this approach can sometimes result in the misinterpretation of amebiasis as other gastroenteritis (GE) conditions. The goal of the work is to produce a machine learning (ML) model that uses laboratory findings and demographic information to automatically predict amebiasis.

Method Data extracted from Jordanian electronic medical records (EMR) between 2020 and 2022 comprised 763 amebic cases and 314 nonamebic cases. Patient demographics, clinical signs, microscopic diagnoses, and leukocyte counts were used to train eight decision tree algorithms and compare their accuracy of predictions. Feature ranking and correlation methods were implemented to enhance the accuracy of classifying amebiasis from other conditions.

Results The primary dependent variables distinguishing amebiasis include the percentage of neutrophils, mucus presence, and the counts of red blood cells (RBCs) and white blood cells (WBCs) in stool samples. Prediction accuracy and precision ranged from 92% to 94.6% when employing decision tree classifiers including decision tree (DT), random forest (RF), XGBoost, AdaBoost, and gradient boosting (GB). However, the optimized RF model demonstrated an area under the curve (AUC) of 98% for detecting amebiasis from laboratory data, utilizing only 300 estimators with a max depth of 20. This study highlights that amebiasis is a significant health concern in Jordan, responsible for 17.22% of all gastroenteritis episodes in this study. Male sex and age were associated with higher incidence of amebiasis ($P = 0.014$), with over 25% of cases occurring in infants and toddlers.

Conclusions The application of ML to EMR can accurately predict amebiasis. This finding significantly contributes to the emerging use of ML as a decision support system in parasitic disease diagnosis.

Keywords Amebiasis, *E. histolytica*, Electronic medical records (EMR), Microscopic diagnosis, Machine learning, Feature selection, Decision tree, Stool RBCs, Leukocytosis, Jordan

Keywords

- The function of keywords is to supplement the information given in the title.
- Words in the title are automatically included in indexes
- keywords serve as additional pointers
- Do not use words or terms in the title as keywords
- 3-10 depending on the journal/conference

e.g.

Keywords: KNN, supervised learning, machine learning

The introduction (1 page)

- 1. Describe the problem
- 2. State your contributions
- ...and that is all

The introduction (e.g.)

1. Introduction

The nearest neighbor approach was first introduced by (Fix & Hodges, 1951) and later studied by (Cover & Hart, 1967). This approach is one of the simplest and oldest methods that used for pattern classification. It often yields efficient performance, and in certain cases, its accuracy is higher than the state-of the-art classifiers (Hamamoto, Uchimura, & Tomita, 1997) (Alpaydin, 1997).

There are two major problems inherited by the design of the KNN (Guo, Wang, Bell, Bi, & Greer, 2003) and (Bhatia & Vandana, 2010) :

1. There is no output trained model to be used, the algorithm has to use all the training examples on each test, therefore its time complexity is linear $O(n)$.
2. Its classification performance depends on choosing the optimal number of neighbors (k), which is different from data sample to another.

The purpose of this study is to solve the second problem, by removing the need for using a specific k with the classifier.

Related work

- **Fallacy** To make my work look good, I have to make other people's work look bad...
- Giving credit to others does not diminish the credit you get from your paper
- Warmly acknowledge people who have helped you
- Be generous to the competition. "In his inspiring paper [Foo98] Foogle shows.... We develop his foundation in the following ways..."
- Acknowledge weaknesses in your approach
- *Foogle, might be your reviewer*

Related work

- Target the most recent studies (good journals)
- You may go back to the history, “the first solution”
- For each reference, briefly describe:
 - Their solution,
 - methods used
 - data used
 - and their results

Related work, e.g.

There are many studies, which have attempted to solve the first problem, depending on reducing the size of the training (Hart, 1968), (Gates, 1972), (Alpaydin, 1997), (Kubat & Jr, 2000) and (Wilson & Martinez, 2000).

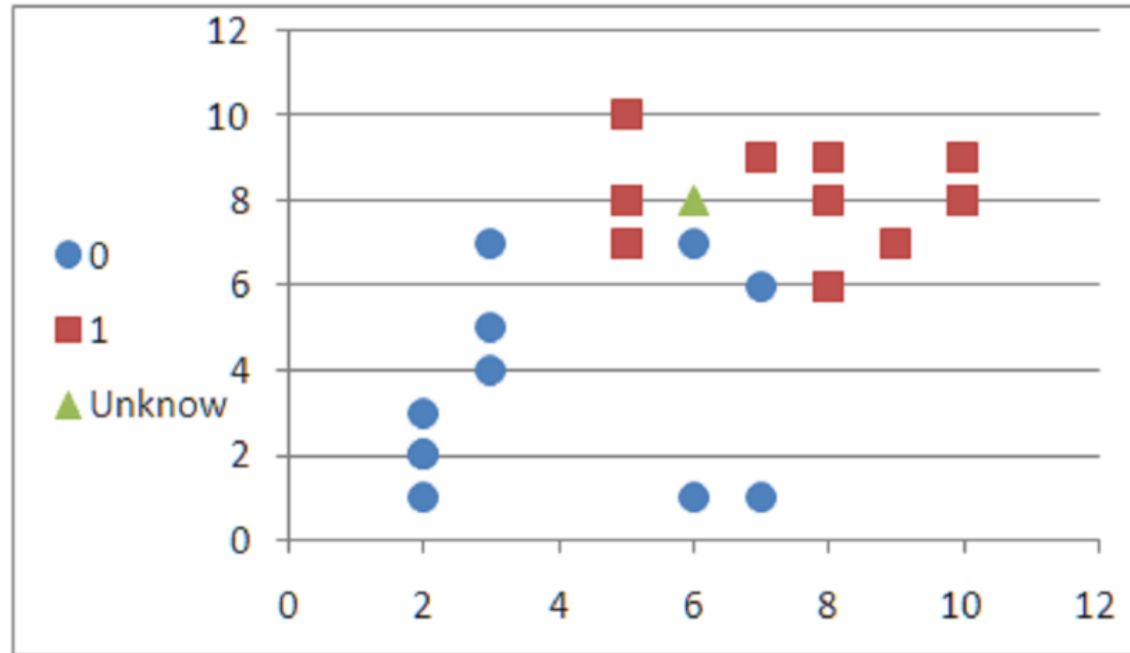
Hart proposed a simple local search method called the “Condensed Nearest Neighbour (CNN)” this method attempts to minimize the number of stored examples and stores only a subset of the training set to be used for classification later. Their idea is based on removing the similar redundant examples (Hart, 1968).

The proposed work

- Define the problem precisely
- Describe your idea thoroughly, use all means possible:
 - Figures
 - Tables
 - Algorithms
 - Examples
 - Equations
- Describe data used
- On the way, cite relevant work in passing

Figures & examples

Refer to the figure in the text, e.g. See Figure 2



class	0	1	1	1	1
order	1	2	3	4	5
weight	1	0.63	0.5	0.43	0.39
k	WS_0	WS_1	Result class=1		
1	1	0			
3	1	1.13			
5	1	1.95			
WS	3	3.08			

Figure 2. Simple example showing the proposed classifier

Algorithms

Algorithm 1: the proposed ensemble KNN classifier

Input: training data set TD, test example TE

Output: class's index

```
1. Array Distances[Size(TD)]
2. index=0
3. For each example as E in TD {
4.     Distances[index]=distanc(E,TE)//any distance function
5.     index=index+1
6. }
7. K=sqrt(Size(TD))
8. Array minClasses[k]
9. minClasses = classes (min k Distances) //ordered by distance
10. Array SW[number of classes in TD]// weight sum for each class
11. Initililze SW// fill with zeros
12. for i=1 to k , i=i+2
13.     for j=1 to i , j=j+1
14.         SW[minClasses[j]]=classes[minClasses[j]]+1/Log(1+j,2)
15. return argmax(classes)
```

Equations

- You may use microsoft equation editor (insert, equation)
- Each equation has a sequence number to the right

$$w(k) = \frac{1}{\log_2(1 + k)} \quad (3)$$

Describe data used

- Sometimes in the results section
- Data can be created by you or a standard database used by other researchers (most common)

Describe data used

*we chose 28 different data sets represent real world classification problems, taken from the UCI Machine Learning Repository (Bache & Lichman, 2014). **Table 1** depicts the data sets used.*

Table 1. Description of the data sets used.

Name	#examples	#features	#classes	data type	Min	Max
Heart	270	25	2	mostly integer	0	564
Balance	625	4	3	pos integer	1	5
Cancer	683	9	2	pos integer	0	9
German	1000	24	2	pos integer	0	184
Liver	345	6	2	pos integer	0	297
Vehicle	846	18	4	pos integer	0	1018

Results and discussion

- Your introduction makes claims
- The body of the paper provides evidence to support each claim
- Check each claim in the introduction, identify the evidence,
- Evidence can be: analysis and comparison, theorems, measurements, case studies, in our case mostly **experiments**
- Describe the experiment protocol, the assumptions, the environment, the data, etc. This is important for **reproducibility** principle.
- Discuss the results, justify why they are good, bad moderate etc.
- Compare results with the others if and only if they used the same data under the same circumstances, otherwise you discuss their results and why they are different.
- Research ethics , **do not ever play here**

Results and discussion

Table 2. The results of the proposed classifier compared to other classifiers, accuracies are the average of 10 runs.

Dataset	1-NN	3-NN	5-NN	7-NN	9-NN	$\sqrt{n}$ -NN	30-NN	45-NN	60-NN	IINC	Proposed
Australian	0.80	0.86	0.87	0.86	0.86	0.86	0.85	0.86	0.86	0.87	0.87
Balance	0.80	0.80	0.83	0.85	0.86	0.88	0.88	0.88	0.88	0.88	0.86
Banknote	1.00	1.00	1.00	1.00	1.00	0.98	0.98	0.97	0.96	1.00	1.00
BCW	0.97	0.97	0.96	0.96	0.96	0.95	0.96	0.96	0.95	0.95	0.96
Cancer	0.96	0.97	0.97	0.96	0.97	0.96	0.96	0.96	0.95	0.95	0.96
Diabetes	0.69	0.72	0.73	0.74	0.75	0.76	0.76	0.76	0.75	0.74	0.74

As can be seen from the results, there is no optimal k , as there is no specific number of neighbors that is suitable for all datasets to be used with nearest neighbor rule. Each data set favor a specific number (k) of neighbors, this note justifies the proposed method, which attempts to use the power of each classifier, and employs it to enhance the overall performance of the proposed method.

A conclusion should be

- stress the importance of the paper
- give the paper a sense of completeness, and
- leave a final impression on the reader.

Conclusion

- Explain what the research has achieved
 - As it relates to the problem stated in the Introduction
 - Revisit the key points in each section
 - Include a summary of the main findings and implications for the field
- Provide benefits and shortcomings of:
 - The solution presented
 - Your research and methodology
- Suggest future areas for research

conclusion e.g.

This work proposes a new classifier based on the KNN classifier, which solves the problem of choosing the number of neighbors that participate in the final decision using the majority rule of the nearest neighbor approach. The proposed method makes use of the ensemble learning approach, where the traditional KNN is used with different number of neighbors each time.

The experimental results using variety of data sets of real world problems have shown superiority of the proposed method over the tradition KNN using variety of k neighbors. In addition, the proposed method was found complete to other classifiers such as the IINC classifier. Moreover, we have shown that the speed of the proposed method (linear time) was found to be better than that of the IINC which is logarithmic linear time.

Future work

- Briefly mention how to further enhance the paper, by using:
 - Different methods
 - more data to further support your claims.
 - Missed aspects
 - Less assumptions
 - Etc.

There is room for enhancing the complexity time of the proposed method using KD-trees (Bentley, 1990) or other hashing technique (Indyk & Motwani, 1998) and (Andoni & Indyk, 2008), this effort is left to be done in the future

References

- We need References for:
 - To distinguish your own ideas from those of someone else.
 - To cite different points of view.
 - To validate what you are writing, by referring to documented evidence. Published work for instance can be used to support your argument and add credibility to your writing.
 - To inform readers of the scope and depth of your reading.

References

- Normally we refer to:
- *Book*: author(s), book title, publisher, date of publication, and page number(s) if appropriate.
- *Journal*: author(s), article title, journal title, date of publication, volume, issue and page number(s).
- *conference*: author(s), article title, conference name, city, date of publication, and page number(s).
- *Newspaper*: author(s), article title, name of newspaper, section title and page number(s) if desired, date of publication.
- *Web site*: author(s), article and publication title where appropriate, as well as a URL, and a date when the site was accessed.
- *Interview*: name of interviewer, interview descriptor (ex. personal interview) and date of interview.

References

- Use a Reference management software for citation, Microsoft word has one
- Go to references, insert citation, add new source

Getting help

- Proofreading
- Get your paper read by as many friends as possible
- Every review is gold dust Be (truly) grateful for criticism as well as praise
- Listen to your reviewers:
 - Read every criticism as a positive suggestion
 - DO NOT respond “ you stupid person..”
 - Thank them warmly. They have given up their time for you.

Publishing

- Choose the journal carefully:
- Make sure the scope of the journal fits your subject
- Do not aim high
- A good journal is either published by a big publisher or indexed in a good indexing service
- Avoid hijacked journals and predatory journals:
<https://scholarlyoa.com/other-pages/hijacked-journals/>

Publishers

- Thompson
- Elsevier
- Springer
- Wiley-Blackwell,
- Taylor & Francis
- Sage
- IEEE
- ACM
- Hindawi
- Any university

Indexing services

- **ISI -Thompson (Impact factor)**
- **Scopus – Elsevier**
- **PubMed**
- Google scholar (h-index)
- Ulrich
- CiteSeer
- The Directory of Open Access Journals (DOAJ)
- ProQuest
- EBSCO*host*
- JSTOR
- Etc.

Impact factor

- A journal must be indexed at ISI Thompson as IF is a trade mark for Thompson only.
- Not every journal has an IF
- 2008 impact factor for an indexed (ISI) journal = A/B

where:

- A = the number of times that all items published in that journal in 2006 and 2007 were cited by indexed publications during 2008.
- B = the total number of "citable items" published by that journal in 2006 and 2007
- The higher the better, however the rejection rate become very high.

Journal finder

- A tool from Elsevier enable research to find the best journal for their research by entering the title, and abstract of their work, the system find the best journal .

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Match	Impact	Editorial Times	Acceptance	Production Times	Open Access	Embargo period	Open Access Fee		User License

Impact Factors

The Ten Most-Cited Journals of 2006

Ranked by total citations tallied in 2006 (the most recent year covered by Thomson Scientific Journal Citation Reports) to previously published articles in each journal.

Rank	Journal	Citations in 2006	Rank for 2005
1	Journal of Biological Chemistry	410,903	1
2	Nature	390,690	2
3	Proc. Natl. Acad. Sci. USA (PNAS)	371,057	3
4	Science	361,389	4
5	J. American Chemical Society	275,769	5
6	Physical Review Letters	268,454	6
7	Physical Review B	212,714	7
8	New England Journal of Medicine	177,505	8
9	Astrophysical Journal	162,136	9
10	Journal of Chemical Physics	157,334	10

Impact Factors

The Ten Most-Cited computer Journals of 2008

	Journal	Papers	Citations	Citations per paper
1	IEEE Journal on Selected Areas in Communication	1,954	23,738	12.15
2	IEEE-ACM Transactions on Networking	951	11,283	11.86
3	IBM Journal of Research and Development	548	6,190	11.30
4	IEEE Transactions on Information Theory	3,466	36,419	10.51
5	ACM Transactions on Graphics	812	7,742	9.53
6	SIAM Journal on Scientific Computing	1,290	11,588	8.98
7	IEEE Transactions on Communications	2,812	23,553	8.38
8	IEEE Transactions on Software Engineering	723	5,707	7.89
9	Computer Methods in Applied Mechanics and Engineering	3,347	26,006	7.77
10	Computer Vision and Image Understanding	773	5,879	7.61

Science Output – Top Ten Countries

Rank	Country	Papers 1998-2008
1	United States	2,798,448
2	Japan	757,586
3	Germany	723,804
4	England	641,768
5	France	517,096
6	People's Republic of China	511,216
7	Canada	388,471
8	Italy	370,053
9	Spain	271,753
10	Russia	262,982

SOURCE: [*Essential Science Indicators*](#)SM from [Thomson Reuters](#)

Science Impact – Top Ten Countries

Rank	Country	Papers 1993-2003	Avg. citations per paper
1	Switzerland	142,982	13.24
2	United States	2,799,593	12.63
3	Netherlands	202,184	11.33
4	Denmark	79,929	11.14
5	Sweden	158,136	10.85
6	Scotland	96,571	10.75
7	England	619,707	10.74
8	Canada	370,928	10.25
9	Finland	74,106	10.17
10	Belgium	103,181	9.74

SOURCE: [*Essential Science Indicators*](#)SM from [Thomson Reuters](#)

The top ten countries ranked according to average citations per paper in all fields (that is, 22 main subject areas, (including general social sciences))

references

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Q & A